01 / FOUNDER DASHBOARD

A credible control spine. Scientific authority remains at zero.

The signed truth is 8 of 29 modules at Phase 0. Four compound modules are milestone movement targets; 12 of 29 is only a conditional outcome after campaign evidence, credentialed evaluation, independent review, and four atomic movements.

8 / 29 LIVE PHASE 0
Signed deterministic controls

4 MOVEMENT TARGETS
All remain pre-Phase-0 today

21 PRE-PHASE-0
Science, data, rights, lab, or external closure

0 PHASE 1 / 2
No module has independent higher-phase validation

● 8 signed Phase 0 ◆ 4 compound movement targets ■ 21 pre-Phase-0 modules

WHAT IS REAL

- **Trust spine.** Typed candidate identity, source custody, intake, authority, packaging, provenance, and red-team controls are signed.
- **Deterministic enforcement.** Malformed, unsupported, tampered, or unauthorized inputs fail closed.
- **Operational shell.** Native dispatch, parallel scoring, lineage, replay, resume/export, and telemetry run.
- **Current data.** 30 rights-known public-corrosion CSVs and six public coating systems are available.

WHAT IS NOT YET PROVEN

- **Scientific model.** No endpoint model or baseline lift has independent validation.
- **Ranking objective.** All 29 modules remain non-ranking; scientific objective count is zero.
- **Laboratory result.** No owned candidate-bound wet-lab result exists; the CRO packet says NOT_EXECUTED.
- **Product claim.** No qualification, procurement, legal, freedom-to-operate, safety, or product-performance authority exists.

FIVE FOUNDER DECISIONS NOW

1 / CREDENTIALS
Authorize distinct external evaluator and reviewer signers.

2 / RELEASE
Fund the four-module campaign → evaluation → review → movement sequence.

3 / COUNTERPARTIES
Contract lab, statistician, chemistry labeling, curation, and counsel.

4 / DATA
Buy owned, candidate-bound measurements with custody and rights.

5 / AUTHORITY
Keep ranking and chemistry-changing mutation disabled until promotion.

AUTHORITY LOCK INTERESTING_FOR_AUDIT is not a rank. RANK: NONE. FINAL AGGREGATE SCORE: NOT AUTHORIZED. Potential novelty is NOT_ESTABLISHED. Unknown proposals are unmade digital proposals. No safety, synthesis, procurement, qualification, performance, or execution claim is authorized.

Authority: live v2 decision report. Green means signed deterministic Phase 0, not scientific validity. Yellow is PRE_PHASE_0 today.

02 / EVIDENCE LADDER + READINESS MODEL

A signed control is not scientific validation—and neither authorizes a product.

Implemented controls prove narrow enforcement. Scientific validation requires locked external evidence. Product authority requires prospective, product-relevant evidence and explicit scope. Current phase comes only from signed movement.

PRE-PHASE 0 / NOT YET AUTHORIZED

Code or artifacts may exist, but required evaluation, review, rights, benchmark, or movement gates remain incomplete.

PHASE 0 / USEFUL CONTROL

Signed validator or enforcement capability inside a narrow applicability domain. Scientific outcome may be NOT_APPLICABLE.

PHASE 1 / INDEPENDENT EVIDENCE

A locked external benchmark passes under independent evaluation and review within a signed domain. Current count: 0.

PHASE 2 / PRODUCT EVIDENCE

Prospective, product-relevant validation and authority beyond a bounded research claim. Current count: 0.

PORTFOLIO MAP / MODULE COUNTS

LANE	TOTAL	● SIGNED	◆ RELEASE	■ BLOCKED
Infrastructure	12	7	0	5
Formulation	8	0	0	8
Performance	4	0	0	4
Molecule	5	1	0	4
TOTAL	29	8	0	21

EVIDENCE CONTRACT

- **Live phase authority.** phase0-decision-report.v2.json
- **Ceilings.** Current 29-module manifest: 14 disabled, 10 audit-only, 5 constraint-only; 0 ranking-eligible.
- **KPIs.** Capability profiles plus hash-bound benchmark artifacts.
- **Plans.** Program work orders and preregistration; always labeled PLANNED NOT EXECUTED.
- **No override.** Stale profile phases and narrative summaries cannot change the live decision.

ENGINEERING EVIDENCE / VALIDATED CONTROLS

30,000 certified red-team negatives

Two byte-identical locked replays; signed evaluation and independent review.

5 suites · 9+ / 44−

Implementation-complete release diagnostics; authority ceilings unchanged.

DATA TRUTH / CURRENT INVENTORY

30
RIGHTS-KNOWN CSVs

6
PUBLIC SYSTEMS

- **Excluded.** Rights-unresolved material is quarantined from ranking and promotion evidence.
- **Owned wet lab.** 0 candidate-bound results.
- **Claim limit.** Local presence is not ownership, permission, or scientific validity.

SCIENTIFIC SIGNALS / NOT VALIDATED

PUBLIC CORROSION

0.666667 = 0.666667 model = baseline · absolute lift 0

CURE EXPLORATORY AUDIT

0.4375 vs 0.375 lift 0.0625 · below locked requirements

Known answers: 13 verified + 1 unverified historical; arithmetic is a distinct 6/6 subset.

Every figure above is classified in the evidence snapshot as validated control, current checked artifact, exploratory result, or inventory fact.

03 / SIGNED TRUST SPINE

Seven controls make evidence tamper-evident, replayable, and claim-bounded.

These are the existing signed Phase-0 infrastructure controls. They establish a deterministic custody and policy spine—not formulation fitness, scientific validity, or product authority.

MODULE	PURPOSE	LIVE STATUS	EVIDENCE / KPI	DECISIVE BLOCKER	NEXT ACTION
amphora-candidate-contract Candidate contract	Enforce typed candidate payloads and canonical identity.	● SIGNED P0 PHASE_0	Signed evaluation + independent review; one live movement event.	None for current deterministic Phase-0 scope.	Preserve hashes and fail-closed contract parity.
amphora-evidence-intake-validator Evidence intake validator	Verify observed bytes, identity, issuer, and signatures.	● SIGNED P0 PHASE_0	Signed fail-closed intake control; no authority upgrade.	None for current deterministic Phase-0 scope.	Maintain signer, revocation, and byte-hash checks.
amphora-model-card-authority Model-card authority	Apply least-privilege authority ceilings to evidence receipts.	● SIGNED P0 PHASE_0	Authority stays audit-only; ranking remains false.	None for current deterministic Phase-0 scope.	Keep policy hashes and signer separation current.
amphora-redteam-audit Red-team audit	Reject tampering, laundering, spoofing, and escalation attacks.	● SIGNED P0 PHASE_0	30,000/30,000 unique negatives; two byte-identical replays.	None for current deterministic Phase-0 scope.	Retain corpus identity and independent signer controls.
amphora-report-packager Report packager	Package claim-locked, evidence-bound decision reports.	● SIGNED P0 PHASE_0	Signed packaging control; legacy claims are not accepted.	None for current deterministic Phase-0 scope.	Preserve report-envelope and claim-lock binding.
amphora-run-provenance Run provenance	Bind replayable run inputs, outputs, lineage, and hashes.	● SIGNED P0 PHASE_0	Replay-bound canonical input/output hashes.	None for current deterministic Phase-0 scope.	Verify replay and event-chain integrity on every run.
amphora-source-ledger Source ledger	Track source identity, citations, rights, and mutation.	● SIGNED P0 PHASE_0	Signed ledger control; unresolved rights stay quarantined.	None for current deterministic Phase-0 scope.	Continue record-level rights and custody checks.

IDENTITY + CUSTODY

Candidate contract, evidence intake, source ledger, and run provenance bind identity, observed bytes, rights context, and replay lineage.

AUTHORITY + ADVERSARIAL CONTROL

Model-card authority and signed red-team audit reject escalation, laundering, spoofing, and out-of-scope use.

FAIL-CLOSED REPORTING

The report packager binds evidence to claims and preserves explicit abstention, negative results, and claim locks.

Signed Phase 0 means useful deterministic control inside a narrow applicability domain. The scientific outcome for these controls is NOT_APPLICABLE.

04 / REMAINING INFRASTRUCTURE + TRANSFER

Five modules still depend on external evidence, rights, or execution.

They are PRE_PHASE_0. Code and fixtures cannot substitute for a laboratory return, an independent benchmark, counsel-defined scope, supplier records, or an explicitly authorized scientific objective.

MODULE	PURPOSE	LIVE STATUS	EVIDENCE / KPI	DECISIVE BLOCKER	NEXT ACTION
amphora-cro-protocol-packager CRO protocol packager	Build candidate-bound external laboratory protocol packets.	■ BLOCKED PRE_PHASE_0	1 positive + 9 negative controls; lab status NOT_EXECUTED.	Live reissue, evaluation, independent review, and movement absent.	Reissue under authorized credentials; verify one movement event.
amphora-optimizer-orchestrator Optimizer orchestrator	Authorize constraints and objectives before selection.	■ BLOCKED PRE_PHASE_0	2 positive + 10 negative controls; zero eligible objectives.	Live reissue, evaluation, independent review, and movement absent.	Promote deterministic shell only; keep objectives disabled.
amphora-ip-fto-guard IP / FTO guard	Surface patent and freedom-to-operate risk without legal conclusions.	■ BLOCKED PRE_PHASE_0	Audit-only; no FTO or legal opinion authority.	Independent counsel and current legal review absent.	Commission counsel review within an explicit jurisdiction and scope.
amphora-known-answer-benchmarks Known-answer benchmarks	Reconcile deterministic controls against known answers.	■ BLOCKED PRE_PHASE_0	13 verified controls + 1 unverified historical; arithmetic is 6/6 subset.	One historical control remains unverified; no 14/14 claim.	Resolve or retire the historical control under independent review.
amphora-sourceability-ehs Sourceability / EHS	Audit sourcing and environment, health, and safety constraints.	■ BLOCKED PRE_PHASE_0	Audit-only outputs; no procurement or EHS approval.	Supplier, procurement, rights, and EHS authority unresolved.	Obtain supplier evidence and independent EHS/procurement sign-off.

TRANSFER PATH / WHERE AUTHORITY STOPS

1 / CANDIDATE-BOUND DRAFT
Protocol packet; execution_authorized: false

→

2 / EXTERNAL EXECUTION
Contract lab, method, custody, deviations

→

3 / SIGNED RETURN
Candidate-bound raw data + uncertainty

→

4 / INDEPENDENT REVIEW
Distinct signer checks method and scope

→

5 / POLICY DECISION
No movement unless every gate passes

CRO PACKAGER
Can bind a protocol to candidate identity. No laboratory result exists; status remains NOT_EXECUTED.

OPTIMIZER
No ranking-eligible module exists. Scientific objectives, optimizer-usable scores, and final aggregate scores remain zero or unauthorized.

RIGHTS + TRANSFER
FTO, sourcing, environment/health/safety, procurement, and known-answer authority require scoped external closure.

Boundary: a packet is not execution; a benchmark fixture is not independent validation; a similarity observation is not novelty or FTO.

05 / FORMULATION SUBMODELS

Arithmetic and boundary controls work. Formulation fitness does not yet exist.

All eight formulation modules. Yellow controls are implementation-complete but remain PRE_PHASE_0; red outputs are audit or constraint-only.

MODULE	PURPOSE	LIVE STATUS	EVIDENCE / KPI	DECISIVE BLOCKER	NEXT ACTION
amphora-formulation-arithmetic Formulation arithmetic	Compute deterministic part and total masses.	■ BLOCKED PRE_PHASE_0	Known answer: 110 g + 40 g = 150 g; 2 positive + 7 negative.	Current reissue lacks its evaluation signature.	Obtain authorized evaluation, independent review, then movement.
amphora-formulation-universe Formulation universe	Account for every Cartesian formulation combination.	■ BLOCKED PRE_PHASE_0	3 × 2 × 2 = 12: 10 solved, 2 declaratively infeasible; 2+ / 9−.	Live reissue, evaluation, independent review, and movement absent.	Run the signed release sequence; preserve declarative-only scope.
amphora-envelope-gates Envelope gates	Enforce signed numeric boundaries and abstention.	■ BLOCKED PRE_PHASE_0	2 positive + 9 negative controls; inclusive margins hit 0 exactly.	Live reissue, evaluation, independent review, and movement absent.	Promote boundary enforcement only, not formulation performance.
amphora-design-rule-score Design-rule score	Check formulation design rules without claiming fitness.	■ BLOCKED PRE_PHASE_0	Audit-only diagnostics; no checked scientific benchmark.	Independent benchmark and review policy incomplete.	Execute the locked internal benchmark with independent review.
amphora-ingredient-library Ingredient library	Resolve cited ingredient constants and applicability.	■ BLOCKED PRE_PHASE_0	Constraint/audit outputs only; no ingredient ranking authority.	Rights and owned supplier evidence remain incomplete.	Bind current supplier sources, rights, units, and review receipts.
amphora-public-corrosion-benchmark Public corrosion benchmark	Benchmark a corrosion endpoint against a fixed baseline.	■ BLOCKED PRE_PHASE_0	Model 0.666667 vs baseline 0.666667; absolute lift 0.	No lift, candidate binding, untouched external systems, or independent validation.	Execute the locked independent public-corrosion campaign.
amphora-solvency-score Solvency score	Estimate solvency compatibility as a bounded constraint.	■ BLOCKED PRE_PHASE_0	Constraint-only output; no optimizer-usable score.	Rights, owned calibration blends, and independent CRO evidence absent.	Run the preregistered blend campaign and locked evaluation.
amphora-tolerance-robustness Tolerance robustness	Test sensitivity to declared formulation tolerances.	■ BLOCKED PRE_PHASE_0	Constraint-only robustness audit; no product tolerance claim.	Owned manufacturing variation and independent validation absent.	Collect candidate-bound process variation data and review.

BOUNDARY OF PROOF

ARITHMETIC / KNOWN ANSWER

110 g + 40 g = 150 g

Proves exact mass accounting only; evaluation signature is absent on the current reissue.

UNIVERSE / KNOWN ANSWER

3 × 2 × 2 = 12

10 solved + 2 declaratively infeasible. “Solved” is accounting, not scientific feasibility.

ENVELOPES / ENFORCEMENT

2+ / 9−

Boundary fixtures prove arithmetic and inclusive-margin enforcement; they do not prove performance.

PUBLIC CORROSION / NO LIFT

Δ = 0

Model and baseline both 0.666667. No candidate binding or independent validation.

Solvency, design-rule, ingredient, and tolerance outputs remain constraint or audit signals. None is an authorized objective.

06 / PERFORMANCE + MOLECULE

Executable diagnostics are not scientific authority.

Four performance and five molecule modules. The only signed module here is the deterministic molecule design gate.

PERFORMANCE / 4 MODULES

MODULE	PURPOSE	LIVE STATUS	EVIDENCE / KPI	DECISIVE BLOCKER	NEXT ACTION
amphora-barrier-corrosion-score Barrier corrosion score	Estimate barrier protection from electrochemical and panel evidence.	■ BLOCKED PRE_PHASE_0	No checked benchmark result; scientific benchmark NOT_EXECUTED.	Rights and 210-panel independent campaign not executed.	Execute 40-system barrier campaign with 10 untouched external systems.
amphora-cure-potlife-gate Cure / pot-life gate	Gate cure and application-life diagnostics within a signed method.	■ BLOCKED PRE_PHASE_0	Exploratory 0.4375 vs 0.375; lift 0.0625; locked gates not met.	Method-comparable CRO execution and independent validation absent.	Run 36 preregistered cure tests and locked evaluation.
amphora-process-qc-twin Process-QC twin	Detect manufacturing process drift and seeded failures.	■ BLOCKED PRE_PHASE_0	Operational campaign only; no Phase-1 attempt claim.	Prospective 30-batch pilot and independent review absent.	Run a separate process-QC operational campaign.
amphora-viscosity-score Viscosity score	Estimate viscosity and sag within declared test conditions.	■ BLOCKED PRE_PHASE_0	No checked scientific benchmark; ranking disabled.	288-cell independent rheology/sag campaign not executed.	Execute locked multi-temperature rheology and sag campaign.

MOLECULE / 5 MODULES

MODULE	PURPOSE	LIVE STATUS	EVIDENCE / KPI	DECISIVE BLOCKER	NEXT ACTION
amphora-molecule-design-gate Molecule design gate	Enforce structural applicability and hard molecule constraints.	● SIGNED P0 PHASE_0	Signed constraint gate; no synthesis or performance claim.	None for current deterministic Phase-0 scope.	Keep structural applicability fail-closed and non-ranking.
amphora-molecule-generator Molecule generator	Propose molecule candidates while preserving lineage and quarantine.	■ BLOCKED PRE_PHASE_0	Mutation changes identity only: QUARANTINED_NO_STRUCTURE_CHANGE.	No validated chemistry-changing operator or synthesis evidence.	Run a separate generator campaign before enabling chemical mutation.
amphora-molecule-novelty Molecule novelty	Measure design-space distance without implying performance.	■ BLOCKED PRE_PHASE_0	Dashboard declaration only; not an authorized objective.	Rights-cleared corpus and independent novelty policy absent.	Lock corpus, labels, distance policy, and independent review.
amphora-molecule-property Molecule property	Compute bounded molecular identity and property diagnostics.	■ BLOCKED PRE_PHASE_0	Executable diagnostics only; no validated scientific authority.	Exact toolkit reproduction and ≥300 eligible structures absent.	Execute the locked molecule-property campaign and independent reconciliation.
amphora-molecule-role-ranker Molecule role ranker	Estimate molecule-role fit from independently labeled examples.	■ BLOCKED PRE_PHASE_0	Role-fit field is a dashboard declaration, not an objective.	Independent chemistry-labeling campaign and benchmark absent.	Acquire blinded role labels and run the locked benchmark.

SEVEN BOUNDED PHASE-1 ATTEMPT CANDIDATES

barrier corrosion · cure / pot-life · molecule property · molecule role · public corrosion · solvency · viscosity

"Attempt" means a preregistered opportunity to produce independent evidence. Completing a campaign does not promise promotion.

SEPARATE OPERATIONAL CAMPAIGNS

molecule generator · process-QC twin

These are operating campaigns, not Phase-1 scientific claims. Generator mutation is not chemistry-changing today.

Cure's 0.0625 exploratory lift is below locked promotion requirements. Public corrosion has zero lift. Both remain unvalidated.

07 / 90-DAY EVIDENCE PROGRAM

The 90-day program is preregistered — and has not been executed.

Spend is justified only if it produces owned, candidate-bound raw evidence with custody, rights, locked analysis, and independent signatures.

PLANNED · NOT EXECUTED No promotion is promised by completing a campaign. Passing the preregistered gates, independent evaluation, review, and signed movement are all still required.

30 / 3	36	288	210 / 40	96 + 32	≥ 300
FORMULATIONS / FAMILIES	PRIMARY CURE RUNS	VISCOSITY / SAG CELLS	PANELS / SYSTEMS	ATTACKS + CONTROLS	ELIGIBLE STRUCTURES

CAMPAIGN DESIGN / LOCKED COUNTS

CAMPAIGN	PLANNED N	DECISIVE DESIGN FEATURE
Formulation coverage	30	Three families; seeded maximin selection without endpoint labels or model ranks.
Cure / application life	36	24 fit runs + 12 sealed primary holdouts; external method comparability.
Viscosity / sag	288 cells	Three temperatures; apparent viscosity, Krebs units, and sag endpoints.
Barrier corrosion	210 panels	40 systems, including 10 untouched external systems; day-56 endpoints.
Independent red team	96 + 32	96 planned attacks plus 32 valid controls across 16 families.
Molecule property	≥ 300	Deduplicated eligible structures; exact toolkit, identity, formula, mass, and rights reconciliation.

COUNTERPARTY DEPENDENCIES

INDEPENDENT EVALUATOR
Executes locked analysis and signs results.

INDEPENDENT REVIEWER
Different trusted signer; checks method and claims.

CONTRACT LABORATORY
Returns raw candidate-bound data and custody records.

STATISTICIAN
Maintains splits, intervals, and preregistered gates.

CHEMISTRY LABELING
Produces blinded role and applicability labels.

DATA CURATION
Resolves identity, units, rights, and deduplication.

COUNSEL
Bounds rights, IP, and freedom-to-operate conclusions.

FOUNDER / OWNER
Protects claim ceilings and stops failed campaigns.

PROMOTION GATE

Raw evidence is complete and rights-eligible; split and benchmark locks match; predefined thresholds pass; evaluator and reviewer are distinct trusted signers; claim locks remain intact; one valid movement event exists.

KILL / HOLD GATE

Stop or hold on rights failure, custody break, split leakage, method mismatch, insufficient sample, baseline non-superiority, confidence interval failure, signer conflict, or any pressure to broaden the applicability domain.

08 / EVOLUTIONARY ENGINE

Deterministic generation can be governed. Selection remains disabled.

The current molecule demo executes a replayable four-submodel path. The compound-campaign contract separately requires ordered requests, retained rejects and duplicates, one typed outcome from every module per card, signed lineage, and no rank.

COMPOUND GENERATION CONTRACT / CURRENT AUTHORITY

1 / LOCKED REQUESTS
Family, parents, role, seed, request ID

→

2 / GENERATION
Deterministic order; retain reject + duplicate

→

3 / 29 OUTCOMES
PASS / FAIL / ABSTAIN / NOT_APPLICABLE

→

4 / POLICY SOLVER
Authority + applicability + coverage

→

5 / NO SELECTION
RANK: NONE; aggregate unauthorized

→

6 / LINEAGE + REPLAY
JCS hashes, event chain, resume, export

OPERATIONAL CAPABILITIES

DETERMINISTIC SEEDS
Repeatable request and run identity.

TYPED ATTEMPTS
Accept, reject, negative, duplicate.

MODULE CONTRACT
Exactly 29 unique outcomes per card.

PROVENANCE
Manifest, inputs, outputs, hashes.

REPLAY + RESUME
Snapshots, event chain, export.

LIVE TELEMETRY
Throughput, workers, status, lineage.

BOUNDED DEMONSTRATION RECEIPT

3	1	2	6	6
POPULATION	GENERATION	WORKERS	ATTEMPTED	VALID

6	0	0	0	0
SCORED	FAILURES	RANKED	OBJECTIVES	SOLUTIONS

EXPLICIT LIMITATIONS / CURRENT

- **Lane.** The bounded demo runs the molecule lane; formulation is inspect-only.
- **Generation.** No locked-grammar unknown exists; chemistry change remains QUARANTINED_NO_STRUCTURE_CHANGE.
- **Outcome.** NO_ELIGIBLE_OBJECTIVES is correct under the signed zero-ranking policy.
- **Dashboard.** Role-fit and novelty fields are declarations, not authorized objectives.

ACTIVATION GATE / FUTURE

- **Scientific promotion.** A named module must independently pass its locked benchmark in a signed domain.
- **Objective authority.** The manifest and model card must explicitly authorize use after promotion.
- **Chemistry mutation.** A structure-changing operator needs validation, quarantine policy, and synthesis constraints.
- **Replay.** Authority hashes, evidence, selection, and lineage must remain reproducible.

OVERALL ENGINE STATUS Yellow operational shell with a red scientific activation gate. The current no-rank outcome demonstrates policy enforcement, not optimization quality.

09 / MVP20 FEEDBACK V5 — INPUT TRUTH

A strong use-case charter. Not yet a scientific dataset.

The adjacent mvp20/feedback/v5 material sharply defines the desired marine-primer outcome and autonomous-engine behavior. It does not contain a catalog, measured candidate rows, a known-solution benchmark, or evidence that a formulation meets the envelope.

EXACT SOURCE PATH	BYTES	SHA-256	CLASSIFIED ROLE
/home/ubuntu/amphora/build/mvp20/feedback/v5/txt.txt	657	57abc1e3dca20849877401d23d46d3af09642a339a63b238549aed0a28f1a2a9	Problem framing
/home/ubuntu/amphora/build/mvp20/feedback/v5/txt1.txt	3,813	1dfdc6f8005bc493d6f0d5433dff1aa2e92c7b1e49594c2acc7671aa8fa699d9	Technical envelope
/home/ubuntu/amphora/build/mvp20/feedback/v5/txt2.txt	8,049	43494f433ca79733cc73d3afcb9c2ab101f506dd51f26f8164eaa10995f842ac	Engine charter

CLASSIFICATION REQUIREMENTS_AND_ENGINE_CHARTER_ONLY · authoritative as founder requirements · not measured scientific evidence

WHAT THE FILES GIVE THE ENGINE

- **A precise first use case.** Marine steel, immersion-service intent, two-component epoxy-polyamide, solvent-borne application.
- **A constraint surface.** 1:1 volume mix; approximately 65% volume solids; VOC, film, cure, storage, and application targets.
- **An operating doctrine.** Self-directed model selection; auditable reasoning; explicit assumptions; empirical feedback.
- **A product narrative.** A clean founder-level definition of why the engine matters and how the loop should work.

WHAT THE FILES DO NOT GIVE THE ENGINE

- **No machine-readable v5 snapshot.** There is no feedback/v5/catalog.json or immutable acquisition/normalization receipt.
- **No truth set.** No labeled known formulations, measured endpoints, custody, rights, uncertainty, or method metadata.
- **No candidate result.** No unknown structure, formulation, campaign replay, CRO result, or independent review.
- **No authority.** The charter cannot establish safety, compliance, qualification, performance, novelty, FTO, procurement, or execution readiness.

FOUNDER INTERPRETATION

The MVP win is not “produce a recipe from three prompts.” It is “turn this envelope into a governed experiment that recognizes known controls, proposes unknowns, and learns from measured outcomes without laundering a score into a product claim.”

Direct source inspected: /home/ubuntu/amphora/build/mvp20/feedback/v5. All three source hashes are recorded in the report evidence snapshot.

10 / KNOWN GENERATION-0 CONTROLS

Known chemistry must be visible, useful, and never mistaken for novelty.

A mixed cohort is the fastest credible demonstration: familiar controls prove identity, group counting, role logic, arithmetic, and abstention before unmade proposals are allowed to look impressive.

EA-P10 / DGEBA-LIKE MODEL

Known-family epoxide anchor. Expected origin: KNOWN_GENERATION_0. Novelty posture: KNOWN_CONTROL_NOT_NOVEL.

EA-P09 / BISPHENOL-F MODEL

Known-family epoxide comparator with a distinct aromatic scaffold context. Not a product recommendation.

EA-P12 / DIETHYLENETRIAMINE

Known polyamine positive-role control for reactive amine-H accounting and equivalent arithmetic.

EA-P19 / IPDA MODEL

Known-family cycloaliphatic diamine model for scaffold and role diversity checks.

HOW A KNOWN CONTROL CAN BE "INTERESTING"

- **Attention, not rank.** INTERESTING_FOR_AUDIT means the record exercises a useful mechanism, scaffold, role, or failure boundary.
- **Expected positive.** The control should pass identity and applicable chemistry checks when the toolchain and ontology are valid.
- **Expected comparison.** Its outcomes establish whether unknowns receive the same treatment and whether the system can recover obvious chemistry.
- **Explicit ceiling.** The flag does not mean novel, superior, compliant, safe, purchasable, qualified, or predicted to meet the coating envelope.

MINIMUM KNOWN-CONTROL RECEIPT

- **Origin.** KNOWN_GENERATION_0 plus source artifact and structure hash.
- **Identity.** Canonical isomeric SMILES, InChIKey, formula, mass, scaffold, packed fingerprint.
- **Chemistry.** Reactive atom indices, oxirane or amine-H count, rejection code, toolkit manifest hash.
- **Traversal.** One typed outcome for all 29 modules; missing science is ABSTAIN or NOT_APPLICABLE.
- **Claim locks.** RANK: NONE and FINAL AGGREGATE SCORE: NOT AUTHORIZED.

THE MVP TEST / KNOWN RECOVERY

TEST	EXPECTED BEHAVIOR	PASS EVIDENCE	FAIL / STOP
Known identity	Recover canonical identity reproducibly	Portable/optimized parity; byte-identical replay	Identity, stereo, formula, mass, or fingerprint drift
Known role	Recognize applicable epoxide or reactive amine control	Signed ontology atom-index evidence	Heuristic-only flag or missing atom indices
Known pairing context	Compute typed equivalent balance without candidate overrides	Finite unit-bound EEW/AHEW calculation	Zero groups, inversion, non-finite input, or override
Known attention flag	Surface as useful control without a rank	Reason code + origin + claim locks	Any novelty, superiority, compliance, or product-performance claim

The EA-P identifiers are pre-campaign audit probes from reports/data/epoxy-amine-20-audit-probes.v1.json. They are not campaign participants, formulations, or validated solutions.

11 / UNKNOWN, UNMADE PROPOSALS

Unknowns run beside controls—not on a privileged path.

The locked ORD grammar is intended to generate unmade structures with reaction lineage. Unknown origin is preserved; potential novelty remains a question until exact structure, literature, patent, rights, and independent-review work is complete.

1 / HOST REQUEST
Family + host-selected parents; no product structure

→

2 / AUTHORIZED GRAMMAR
Extraction-only ORD lineage and signed hashes

→

3 / NATIVE CHEMISTRY
Typed RDKit facts, rejects, duplicates, atom evidence

→

4 / MIXED COHORT
Known controls + unknowns; one traversal policy

WHAT MAKES AN UNKNOWN WORTH INSPECTING

- **Valid chemistry record.** Single-component, supported valence/elements, stable canonical identity, complete lineage.
- **Role evidence.** Signed ontology finds relevant atoms; molecule facts support a resin or hardener audit context.
- **Coverage value.** Adds scaffold or mechanism coverage to the bounded 20-card/12-pair audit cohort.
- **Transparent uncertainty.** Every missing endpoint, source, safety, process, or performance fact remains explicit.

WHAT "POTENTIALLY NOVEL" MUST MEAN

- **Permitted wording.** Potential novelty is NOT_ESTABLISHED; review follow-up only.
- **Required checks.** Structure deduplication, source lineage, literature and patent searches, date-bounded evidence, and counsel-defined FTO scope.
- **Separation.** Novelty observations cannot become an aggregate score or override chemistry, rights, safety, or evidence gates.
- **Current truth.** No locked-grammar unknown has been generated here because ORD objects, toolchains, rights receipts, and signatures remain incomplete.

TWO CONCEPTUAL NOVELTY-REVIEW EXAMPLES / REQUEST LINEAGE ONLY

REQUEST	AUTHORIZED FAMILY	EXACT HOST PARENTS	ROLE	CURRENT POSTURE
resin-12	ORD_GLYCIDYL_ETHER_FORMATION	Oc1ccc(S(=O)(=O)c2ccc(O)cc2)cc1 + ClCC1CO1	resin	UNMADE · NOT_ESTABLISHED · AUDIT_ONLY
hardener-07	ORD_REDUCTIVE_AMINATION	NCc1ccccc1CN + CC=O	hardener	UNMADE · NOT_ESTABLISHED · AUDIT_ONLY

CURRENT GATE Deterministic seed 20260719 and the request records are charter inputs. No product structure, ORD reaction/dataset lineage, campaign card, or measured result exists. The examples do not invent campaign structures and authorize no safety, synthesis, execution, novelty, or performance claim.

12 / CENTRAL SOLVER + ADAPTIVE LOOP

Many models emit typed evidence. One central policy solver decides what may influence selection.

The solver does not average 29 scores. It composes PASS, FAIL, ABSTAIN, and NOT_APPLICABLE outcomes under authority, applicability, evidence-strength, and coverage rules. Adaptation begins only from a signed, candidate-bound empirical return.

MIXED GENERATION 0

Known controls + locked unknowns + explicit rejects and duplicates

→

29 MODULE OUTCOMES

PASS / FAIL / ABSTAIN / NOT_APPLICABLE; scalar only when typed and unit-bound

→

CENTRAL POLICY SOLVER

Authority, applicability, evidence strength, coverage, claim locks, and objective checks

↓

12 AUDIT PAIRS

Deterministic coverage; typed equivalent balances; no formulation-performance claim

→

CRO FEASIBILITY DRAFTS

Controls, custody, immutable holds; execution_authorized: false

→

SIGNED MEASURED RETURN

Candidate-bound observations, method, uncertainty, deviations, rights, and distinct signatures

WHAT ADAPTS

- **Calibration.** Within a locked endpoint and applicability domain, predictions can be recalibrated from measured residuals.
- **Uncertainty.** Abstention regions and confidence bounds can tighten or widen from observed coverage.
- **Experimental allocation.** After authority exists, the next round can emphasize information gain while retaining controls.
- **Method status.** A model can be held, revised, or promoted only through preregistered evaluation and independent review.

WHAT DOES NOT ADAPT SILENTLY

- **Claim locks.** No feedback can enable rank, procurement, synthesis, safety, qualification, or performance authority.
- **Truth labels.** Known controls never become “novel”; failed experiments never disappear.
- **Splits.** Extraction, validation, and held-out partitions remain hash-locked; held-out data never enters training or PGO.
- **Methods.** No automatic model swap, threshold weakening, candidate override, or unreviewed objective change.

INITIAL MVP WIN

Recover known controls, generate valid unknowns, explain why each is surfaced, package a replayable audit cohort, and ingest one independently measured return—without producing an unauthorized rank.

13 / WHAT IS STRONG, WHAT IS WEAK, HOW IT CLOSES

The control architecture is credible. The compound science is not yet demonstrated.

This distinction is the core founder message: engineering trust is an asset, but it becomes a product only when real chemistry, rights, replay, and measured feedback close the loop.

STRONG NOW

- **Fail-closed authority.** Eight signed Phase-0 controls keep ranking and product claims disabled.
- **Deterministic contracts.** JCS v2 hashing, exact artifact bindings, typed module outcomes, and separate scientific/execution receipts are implemented foundations.
- **Native ORD foundation.** Pinned acquisition verifies 550 object hashes/sizes, gzip integrity, and explicit Parquet exclusion; vendored protobuf codegen removes host protoc assumptions.
- **Toolchain contract.** Portable and optimized RDKit profiles are separately hash-bound with ABI, CPU, SBOM, and parity requirements.
- **Honest reporting.** The final verified report path is protected from fixture chemistry and premature success claims.

WEAK / BLOCKED NOW

- **Input evidence.** mvp20/feedback/v5 is three narrative files, not a machine-readable benchmark or measured corpus.
- **Chemistry runtime.** No verified portable/optimized RDKit installation exists in artifacts/toolchains.
- **Campaign evidence.** No acquired ORD objects, split, real grammar, generated unknowns, 12 pairs, 20 cards, or campaign replay exists.
- **External trust.** Rights receipts, evaluator signature, sealed attack suite, independent review, and replay receipt are absent.
- **Product loop.** No candidate-bound wet-lab measurement proves adaptation or any marine-coating endpoint.

CLOSURE PLAN / EVIDENCE, NOT RHETORIC

ORDER / OWNER	CLOSE	PASS RECEIPT	FOUNDER VALUE	STOP CONDITION
1 / Platform	Build and verify both native RDKit profiles	Tests, ABI closure, SBOM, provenance, reproducible portable trees	Trusted chemistry identity and performance baseline	Any source, ABI, test, CPU, Python, or parity mismatch
2 / Data + chemistry	Acquire/decode/split ORD and extract two-family grammar	550 objects; complete reaction/exclusion accounting; zero leakage; real lineage	Unknown generation grounded in public reaction evidence	Missing object, corrupt protobuf, leakage, or unsupported family
3 / Data + science	Materialize real feedback/v5 benchmark	300-structure truth cohort, raw PubChem responses, signed ChEBI/PubChem rights	Known-control recovery has a measurable acceptance test	Rights, identity, count, or locked threshold failure
4 / Science + assurance	Run mixed known/unknown campaign + red team	At least 24 valid structures; 12 pairs; 20 cards; sealed attacks; replay	Founder-demonstrable compound MVP	Silent acceptance, rank injection, scientific drift, swap, or memory failure
5 / Lab + evaluator	Return one independent measured loop	Signed candidate-bound measurement, method/custody/uncertainty, clean import and recalibration	First proof that the engine can learn from reality	Custody, method, rights, or independent-review failure

No threshold is weakened and no fixture is substituted if a closure step fails. The failure remains a negative result.

14 / FOUNDER ACTIONS + DECISION GATES

Spend should purchase independent evidence and owned data.

Do not fund additional unvalidated ranking logic while the existing authority, data, and external-evidence gates remain open.

0–30 DAYS / RELEASE + CONTRACT

- **Credentials.** Obtain authorized external evaluator and reviewer credentials.
- **Four targets.** Complete campaign → evaluation → independent review → movement for property, generator, arithmetic, and CRO packaging.
- **Trajectory.** Verify one event per module; projected 12/29 only after success, not current.
- **Counterparties.** Contract lab, statistician, labeling, curation, and counsel.

31–90 DAYS / EXECUTE

- **Laboratory.** Run cure, rheology/sag, solvency, and barrier campaigns.
- **Data.** Execute public-corrosion, molecule-property, and role-label work orders.
- **Custody.** Collect raw candidate-bound evidence with identity, rights, and source lineage.
- **Stops.** Enforce preregistered kill gates without scope broadening.

90+ DAYS / DECIDE

- **Benchmarks.** Run locked independent evaluations and distinct-signer reviews.
- **Promotion.** Move only passing modules inside signed applicability domains.
- **Objectives.** Activate ranking only after explicit module authority exists.
- **Mutation.** Enable chemistry change only after validated operator and safety gates.

FOUNDER DECISION TABLE

DECISION	OWNER	DEPENDENCY	COST	SUCCESS SIGNAL	KILL CRITERION	AUTHORITY GAINED
Authorize four-module compound release	Founder / security	Distinct trusted evaluator + reviewer	LOW	Four valid evaluations, reviews, and one movement each	Signer conflict, hash drift, or claim-lock change	Projected 12/29 Phase 0 — only after movement
Contract coatings laboratory	Science lead	Method-comparable lab + statistician	HIGH	Complete raw cure, rheology, solvency, and barrier evidence	Custody break, rights failure, method mismatch, or sample shortfall	No automatic gain; evidence eligible for locked evaluation
Acquire molecule evidence	Chemistry lead	Data curator, exact toolkit reproducer, labelers	MED	≥300 eligible property structures + blinded role labels	Identity mismatch, rights gap, leakage, or toolkit drift	No automatic gain; bounded Phase-1 attempt evidence
Commission rights / legal closure	Operations + counsel	Supplier, procurement, environment/health/safety and IP records	MED	Signed, scoped rights and legal conclusions	Unbounded opinion, stale sources, or unresolved jurisdiction	Audit / decision authority only within signed scope
Hold ranking activation	Founder / model authority	Passing independent benchmark + updated signed manifest	LOW	At least one explicitly authorized objective in a signed domain	Any pressure to infer authority from code, tests, or demo	Objective use only for the promoted module and domain

EXACT MVP ACCEPTANCE SIGNAL

One hash-bound replay recovers all four known controls; records at least one valid unmade proposal from each authorized reaction family with complete lineage and 29 typed outcomes; retains rejects and duplicates; ingests one independently measured, signed, candidate-bound return; and emits RANK: NONE with FINAL AGGREGATE SCORE: NOT AUTHORIZED.

KILL / HOLD CRITERIA

Stop on toolchain or replay mismatch, known-control miss, unsupported lineage, missing family coverage, rights or custody failure, held-out leakage, signer conflict, baseline non-superiority, hidden rank, aggregate-score injection, or pressure to broaden the signed domain.

Funding rule: purchase independent evidence and owned data—not additional unvalidated ranking logic. No promotion, product claim, or chemistry-changing mutation without the exact signed gate.